

Xuanyu Peng

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BIO

I'm Xuanyu Peng, a fourth-year Ph.D. student in computer science at the University of California, San Diego, advised by [Prof. Loris D'Antoni](#). My research aims to improve the reliability of software systems through program language techniques, with a current focus on automatically constructing correct, optimizing, and customizable (tensor) compilers. I obtained my B.S. in Computer Science from Turing Class, Peking University, where I worked with [Prof. Yingfei Xiong](#) and [Prof. Di Wang](#).

EDUCATION

University of California, San Diego

La Jolla, CA, USA

Ph.D. in Computer Science

Sep 2023 - Present

Began at University of Wisconsin–Madison (Sep 2023 - Aug 2024); transferred to UCSD in Sep 2024

- Synthesizing Static Analysis: Developed **NiceToMeetYou**, a program synthesis framework for automating abstract interpretation in dataflow analyses heavily used by production compilers.
- Specification Inference: Built **Loud**, a specification inference framework that extracts sound and precise properties from programs.

Peking University

Beijing, China

Bachelor of Science in Computer Science & Technology, Turing Class

Sep 2019 - June 2023

GPA: **3.80**/4.0 (rank: **14th**/200)

WORK EXPERIENCE

AWS Annapurna ML

Jun 2026 – Sep 2026

Applied Scientist Intern, mentored by Chungha Sung

- Automatic NKI Kernel Optimization: Developed a synthesis-based approach to automatically generate and fuse NKI kernels, improving vLLM model serving performance on AWS Trainium.

AWS Cloud Operations

Jun 2025 – Sep 2025

Applied Scientist Intern, mentored by Victor Nicolet and Joey Dodds

- Data Transformation Synthesis: Proposed a schema-guided approach for synthesizing JSON transformations, applied to automatic generation of cloud scripts and cloud configuration updates.

PUBLICATIONS

- [1] Dominic Kennedy*, Xuanyu Peng*, Yuyou Fan, John Regehr, Loris D'Antoni, and Ben Greenman. "A Fistful of Abstract Values: Synthesizing the Transformers Compilers Actually Need". In: *POPL* (2027). To appear.
- [2] Xuanyu Peng*, Dominic Kennedy*, Yuyou Fan, Ben Greenman, John Regehr, and Loris D'Antoni. "Nice to Meet You: Synthesizing Practical MLIR Abstract Transformers". In: *POPL* (2026).
- [3] Kanghee Park*, Xuanyu Peng*, and Loris D'Antoni. "LOUD: Synthesizing Strongest and Weakest Specifications". In: *OOPSLA* (2025).
- [4] Yican Sun, Xuanyu Peng, and Yingfei Xiong. "Synthesizing Efficient Memoization Algorithms". In: *OOPSLA* (2023).

* denotes equal contribution.

SCHOLARSHIPS AND AWARDS

John Hopcroft's Scholarship at Peking University

2022

Ubiquant Scholarship

2021

Peking University Merit Student

2020

Gold Award (7th place), The 44th ACM-ICPC Asia Regionals Xuzhou Site

2019

Silver Award, The 5th CCPC Xiamen Site

2019